

REMARKS ON PRIMITIVE REGULATION

Milan Rosko
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ORCID: 0009-0003-1363-7158

Abstract

We prove, and mechanize in ROCQ, an obstruction to closure-level EXCLUDED MIDDLE for primitive regulators $C : \mathbf{Form} \rightarrow \mathbf{Prop}$ over the closed implication–falsity fragment $A, B ::= \perp \mid A \rightarrow B$. Write $\mathbf{LEM}(C)$ for the demand that $C(A) \vee C(\neg A)$ hold for every formula A . If C is closed under MODUS PONENS, is consistent, and admits a formula B satisfying $B \simeq_C \neg B$, where $A \simeq_C B$ abbreviates $C(A \rightarrow B) \wedge C(B \rightarrow A)$, then $\mathbf{LEM}(C)$ is impossible. In fact, consistency excludes both $C(B)$ and $C(\neg B)$, so the global conclusion uses only the excluded-middle instance at B . A goal-restricted anti-diagonal premise suffices to produce such a point.

1 INTRODUCTION

True or false: *True or false* holds for every formula.

This question already *pressures* the very form of answer under examination. To answer internally with *yes* is to accept unrestricted closure-level EXCLUDED MIDDLE. To answer internally with *no* is itself a closure-level verdict about the totality principle.

1.1 BACKGROUND

Diagonal limitations recur across several formal settings. RUSSELL’S ANTI-NOMY [Russell, 1908; Whitehead and Russell, 1927] exposes unrestricted comprehension; INCOMPLETENESS [Gödel, 1931; Rosser, 1936] turns an arithmetized representation of syntax back upon derivability; UNDEFINABILITY [Tarski, 1933] turns DIAGONALIZATION against a proposed truth predicate; computability-theoretic limitations [Church, 1936; Turing, 1937; Kleene, 1952; Rice, 1953] combine effective indexing with self-application; and LÖB’S THEOREM [Löb, 1955] controls self-referential provability through derivability conditions. Their objects and conclusions differ, but each exposes a limit on a system that is expressive enough to classify its own indexed behaviour.

The present paper isolates this *pressure* at the level of primitive regulation. Its question is whether a formula-indexed acceptance predicate can simultaneously support self-applicable representation, closure under detachment, consistency, and EXCLUDED MIDDLE at the level of accepted formulas. The resulting obstruction arises already in the closed implication–falsity fragment.

1.2 INTUITION

Prior to the formal definitions, let C denote a primitive acceptance predicate

$$\text{Regulator } C, \tag{1}$$

on the closed implication–falsity fragment. Thus $C(A)$ means that the formula A is accepted. Four demands organize the argument. First, a supplied evaluation interface is self-applicable and represents the single anti-diagonal behaviour needed at $\perp$. Second, C supports *detachment*: if $C(A \rightarrow D)$ and $C(A)$, then $C(D)$. Third, C is *consistent*: it does not accept $\perp$. Fourth, C satisfies closure-level EXCLUDED MIDDLE: for every formula A , either $C(A)$ or $C(\neg A)$.

The obstruction has three stages. Goal-restricted representation first produces a formula B satisfying

$$B \simeq_C \neg B, \quad (2)$$

so that C accepts both $B \rightarrow \neg B$ and $\neg B \rightarrow B$. Closure-level EXCLUDED MIDDLE then supplies one of the two acceptance facts

$$C(B) \quad \text{or} \quad C(\neg B). \quad (3)$$

Whichever branch is supplied, detachment along the accepted fixed-point implications produces both $C(B)$ and $C(\neg B)$, and hence $C(\perp)$. Consistency turns this collapse into a contradiction.

The global principle is therefore obstructed at a single formula. The result is stronger than the bare failure of a universal schema: once the diagonal point is present, one instance of closure-level EXCLUDED MIDDLE already carries the entire contradiction. The same local pressure later separates signed classification from ordinary membership decision and from negative-only refutation.

1.3 CONTRIBUTIONS

In the companion ROCQ development, **M001** and **L001** are module identifiers: **M001** supplies the executable syntax and proof-checking substrate, while **L001** supplies the abstract closure-obstruction layer over that syntax.

We make five contributions. (i) We derive the failure of closure-level EXCLUDED MIDDLE for every consistent, modus-ponens-closed primitive regulator admitting a closure-equivalence negation fixed point, and show that the proof uses only the excluded-middle instance at that point, the fixed-point direction aligned with the supplied branch, and the corresponding detachment instances. (ii) We isolate the goal-restricted anti-diagonal premise sufficient to produce the fixed point. (iii) We separate signed classification from ordinary membership decision and negative-only refutation. (iv) We prove the four local hypotheses sharp by explicit elementary acceptance predicates. (v) We connect the abstract theorem to the checked-derivability predicate supplied by **M001** and verify the proof-critical contracts in ROCQ.

1.4 ROADMAP

The exposition follows this division. Section 2 develops the **M001** substrate by fixing the closed $\perp / \rightarrow$ syntax, the closure vocabulary, and the checked-derivability instance. Section 3 develops the **L001** obstruction layer by proving the local collapse, deriving the obstruction to closure-level EXCLUDED MIDDLE, stating the exact goal-restricted premise sufficient for diagonal production, developing the classification consequences, and establishing

sharpness of the four local hypotheses. Section 4 records the **M001** syntactic substrate and the six certified **L001** contracts. Section 5 explains why the global obstruction is already determined by one local branch.

2 SUBSTRATE

2.1 OBJECT SYNTAX

Definition 2.1.1 (Syntax). Working in a constructive metatheory [Troelstra and van Dalen, 1988], such as the CALCULUS OF CONSTRUCTIONS [Coquand and Huet, 1988], we take as our object language a minimal propositional language containing only falsity and implication.

$$A, B ::= \perp \quad | \quad A \rightarrow B, \quad \neg A \triangleq A \rightarrow \perp. \quad (4)$$

We write **Form** for the type of formulas; formulas are finite trees over $\{\perp, \rightarrow\}$. There are no atoms, propositional variables, quotation constructors, object-level substitution operations, or semantic truth predicates in this language.

Lemma 2.1.2 (No formula is its own negation). For every formula,

$$\forall A : \text{Form}, A \neq A \rightarrow \perp. \quad (5)$$

Proof. $\neg A = A \rightarrow \perp$ has A as a proper subterm. A finite tree cannot be a proper subterm of itself. ■

Remark 2.1.3. Lemma 2.1.2 forces every fixed-point claim below to be stated via $\simeq_C$, never as a syntactic identity of formulas.

2.2 CLOSURE PREDICATES

Definition 2.2.1 (Primitive regulator with closure predicate). A *primitive regulator* on the formula language is a predicate

$$C : \text{Form} \rightarrow \text{Prop}. \quad (6)$$

We write $C(A)$, read “ C accepts A .” Primitive regulation is the bare acceptance layer: no semantic truth, model satisfaction, arithmetic provability, or model of computation is implied, and $C(A)$ is treated as a mere proposition. We also call C a *closure predicate* when emphasizing closure properties. No deduction, cut, weakening, substitution, or detachment law is implicit in the name; every rule used below is an explicit hypothesis. Every later result is parametric in C and never inspects its definition.

Remark 2.2.2. Throughout, omitted parameters in $\text{LEM}(C)$, $\text{MP}(C)$, $\text{Det}_C(A, D)$, $\text{Cons}(C)$, $\text{LocalLEM}_C(B)$, $\text{Signed}(C)$, $\text{MemDec}(C)$, $\text{RefComp}(C)$, $\text{Ref}(C)$, and $\text{NegFP}_C(B)$ refer to the ambient closure predicate C .

Definition 2.2.3 (Closure equivalence). For formulas A and B , closure equivalence means mutual acceptance of the corresponding implications. We define

$$A \simeq_C B \triangleq C(A \rightarrow B) \wedge C(B \rightarrow A). \quad (7)$$

Thus A and B are equivalent relative to C when C accepts both directions between them.

Remark 2.2.4. $\simeq_C$ records exactly the two accepted implications between A and B . The name does not endow a bare C with reflexivity, transitivity, congruence, or substitution. By Lemma 2.1.2, $A = \neg A$ never holds, whereas $A \simeq_C \neg A$ may, depending on C .

Definition 2.2.5 (Closure-equivalence negation fixed point). A formula $B : \text{Form}$ is a *closure-equivalence negation fixed point* for C when

$$\text{NegFP}_C(B) \triangleq B \simeq_C \neg B. \quad (8)$$

Equivalently,

$$\text{NegFP}_C(B) \triangleq C(B \rightarrow \neg B) \wedge C(\neg B \rightarrow B). \quad (9)$$

This is the fixed-point hypothesis used by the obstruction theorem.

Definition 2.2.6 (MODUS PONENS and consistency). The structural properties of C are

$$\begin{aligned} \text{MP}(C) &\triangleq \forall A, B : \text{Form}, (C(A \rightarrow B) \wedge C(A)) \rightarrow C(B), \\ \text{Cons}(C) &\triangleq C(\perp) \rightarrow \perp. \end{aligned} \quad (10)$$

Definition 2.2.7 (Local detachment). For formulas A, D , the corresponding detachment instance is

$$\text{Det}_C(A, D) \triangleq C(A \rightarrow D) \rightarrow C(A) \rightarrow C(D). \quad (11)$$

Thus $\text{MP}(C)$ entails $\text{Det}_C(A, D)$ for every A, D , while a local argument may name only the instances it uses.

Definition 2.2.8 (Closure-level EXCLUDED MIDDLE). Closure-level EXCLUDED MIDDLE for C is the totality principle

$$\text{LEM}(C) \triangleq \forall A : \text{Form}, C(A) \vee C(\neg A). \quad (12)$$

It requires the regulator to accept a positive or negative side for every object formula. The principle is a property of C ; the obstruction to it below is proved in the constructive metatheory.

Definition 2.2.9 (Local closure-level EXCLUDED MIDDLE). For a fixed formula B , local closure-level EXCLUDED MIDDLE at B is the single instance

$$\text{LocalLEM}_C(B) \triangleq C(B) \vee C(\neg B). \quad (13)$$

Clearly, $\text{LEM}(C)$ entails $\text{LocalLEM}_C(B)$ for every B . The local obstruction below assumes only this one instance.

Definition 2.2.10 (Signed classification, membership decision, and refutation). A *signed classification* for C is a Boolean classifier whose two verdicts carry internal closure certificates:

$$\text{Signed}(C) \triangleq \exists s : \text{Form} \rightarrow \{\text{tt}, \text{ff}\}, \quad \begin{aligned} &\forall A : \text{Form}, s(A) = \text{tt} \rightarrow C(A), \\ &\forall A : \text{Form}, s(A) = \text{ff} \rightarrow C(\neg A). \end{aligned} \quad (14)$$

An *ordinary membership decision* instead decides the metatheoretic proposition $C(A)$:

$$\begin{aligned} \text{MemDec}(C) &\triangleq \exists m : \text{Form} \rightarrow \{\text{tt}, \text{ff}\}, \\ \forall A : \text{Form}, m(A) = \text{tt} &\longleftrightarrow C(A). \end{aligned} \quad (15)$$

The additional bridge from non-membership to an accepted object-language negation is

$$\text{RefComp}(C) \triangleq \forall A : \text{Form}, \neg C(A) \rightarrow C(\neg A). \quad (16)$$

A *refutation* is the negative-only counterpart, with no constraint on the false branch:

$$\text{Ref}(C) \triangleq \exists r : \text{Form} \rightarrow \{\text{tt}, \text{ff}\}, \quad \forall A : \text{Form}, r(A) = \text{tt} \rightarrow C(\neg A). \quad (17)$$

Remark 2.2.11. Signed classification and membership decision are intentionally distinct. A false signed verdict already certifies $C(\neg A)$; a false membership verdict supplies only $\neg C(A)$. The latter reaches an accepted negation only through $\text{RefComp}(C)$. Since no model of computation is fixed, none of these abstract conditions alone states recursion-theoretic undecidability.

Definition 2.2.12 (REGULATOR THEORY). The mechanized substrate also provides a concrete syntactic instance. A REGULATOR THEORY is a pair

$$R \triangleq (\pi, T), \quad (18)$$

where π selects either the minimal K/S profile or the K/S profile extended with the explicit EFQ schema, and T is a Boolean axiom-membership function over object formulas. The interface need not carry a finite enumeration; the mechanization provides `FiniteAxiomSet` separately when the axiom source itself must be finite data. A context Γ supplies local assumptions but is not part of the theory. Together, R and Γ induce the closure predicate

$$C_{R,\Gamma}(A) \triangleq \Gamma \vdash_R A, \quad (19)$$

where $\vdash_R$ denotes checked derivability in R . This gives one family of closure predicates to which the abstract vocabulary may be applied. The obstruction theorem itself remains parametric in C and does not inspect the checker or assume that every C arises from a REGULATOR THEORY.

Proposition 2.2.13 (Checked derivability is closed under detachment). For every REGULATOR THEORY R and context Γ , the induced closure predicate satisfies

$$\text{MP}(C_{R,\Gamma}). \quad (20)$$

Proof. Suppose $C_{R,\Gamma}(A \rightarrow D)$ and $C_{R,\Gamma}(A)$. These are checked derivations of $A \rightarrow D$ and A in the same theory and context. The M001 theorem `regulator_theory_checked_derivable_mp_lemma` composes their finite checked proof scripts into a checked derivation of D , which is $C_{R,\Gamma}(D)$. ■

3 OBSTRUCTION

3.1 LOCAL COLLAPSE

Theorem 3.1.1 (Branchwise goal-relative collapse). Let $B, G : \text{Form}$. Assume

$$\text{Det}_C(B, B \rightarrow G), \quad \text{Det}_C(B \rightarrow G, B), \quad \text{Det}_C(B, G), \quad (21)$$

and

$$(C(B) \wedge C(B \rightarrow (B \rightarrow G))) \vee (C(B \rightarrow G) \wedge C((B \rightarrow G) \rightarrow B)). \quad (22)$$

Then

$$C(G). \quad (23)$$

Proof. In the first branch, $\text{Det}_C(B, B \rightarrow G)$ gives $C(B \rightarrow G)$, and $\text{Det}_C(B, G)$ then gives $C(G)$. In the second branch, $\text{Det}_C(B \rightarrow G, B)$ gives $C(B)$, after which $\text{Det}_C(B, G)$ again gives $C(G)$. Thus each branch uses only its corresponding fixed-point implication, one branch-specific detachment instance, and the final instance $\text{Det}_C(B, G)$. ■

Theorem 3.1.2 (Goal-relative branch collapse). Let $B, G : \text{Form}$. Assume $\text{MP}(C)$,

$$B \simeq_C B \rightarrow G, \quad (24)$$

and the local branch

$$C(B) \vee C(B \rightarrow G). \quad (25)$$

Then

$$C(G). \quad (26)$$

Proof. The hypothesis $\text{MP}(C)$ supplies the three local detachment instances in Theorem 3.1.1, while the closure equivalence supplies both oriented implications. If $C(B)$ holds, use the first branch of that theorem; if $C(B \rightarrow G)$ holds, use the second. ■

Theorem 3.1.3 (Local branch collapse). Assume $\text{MP}(C)$, $\text{NegFP}_C(B)$, and $\text{LocalLEM}_C(B)$. Then

$$C(\perp). \quad (27)$$

Proof. Apply Theorem 3.1.2 with $G := \perp$. Since $\neg B$ abbreviates $B \rightarrow \perp$, its closure-equivalence and local-branch hypotheses are exactly $\text{NegFP}_C(B)$ and $\text{LocalLEM}_C(B)$. ■

Theorem 3.1.4 (Fixed-point gap). Assume $\text{MP}(C)$, $\text{Cons}(C)$, and $\text{NegFP}_C(B)$. Then

$$\neg C(B) \wedge \neg C(\neg B). \quad (28)$$

Proof. If $C(B)$ held, it would supply the left side of $\text{LocalLEM}_C(B)$; Theorem 3.1.3 would then give $C(\perp)$, contrary to $\text{Cons}(C)$. The same argument from $C(\neg B)$ uses the right side of $\text{LocalLEM}_C(B)$. Hence neither branch is accepted. ■

Theorem 3.1.5 (Core diagonal obstruction). Assume $\text{MP}(C)$, $\text{Cons}(C)$, $\text{NegFP}_C(B)$, and $\text{LocalLEM}_C(B)$. Then

$$\perp. \quad (29)$$

Proof. Theorem 3.1.4 gives $\neg C(B)$ and $\neg C(\neg B)$. Either side of $\text{LocalLEM}_C(B)$ therefore yields $\perp$. ■

Remark 3.1.6. This theorem is the primitive obstruction. It does not mention the representability premise or any uniform bivalence schema. A closure-equivalence negation fixed point, detachment, consistency, and local bivalence at the fixed point are incompatible.

Corollary 3.1.7 (Closure-level EXCLUDED MIDDLE fails at a negation fixed point). Assume $\text{MP}(C)$, $\text{Cons}(C)$, and $\text{NegFP}_C(B)$. Then

$$\neg \text{LEM}(C). \quad (30)$$

Proof. Suppose $\text{LEM}(C)$ holds. Specializing it at B gives $\text{LocalLEM}_C(B)$. Theorem 3.1.5 then yields $\perp$. ■

Corollary 3.1.8 (Checked-regulator obstruction). For every REGULATOR THEORY R , context Γ , and formula B , if $C_{R,\Gamma}$ is consistent and B is a closure-equivalence negation fixed point for $C_{R,\Gamma}$, then

$$\neg \text{LEM}(C_{R,\Gamma}). \quad (31)$$

Proof. Proposition 2.2.13 supplies $\text{MP}(C_{R,\Gamma})$. Corollary 3.1.7 applies to the induced closure predicate. ■

Neither M001 nor L001 constructs an instance of $\text{Eval}_\perp(C_{R,\Gamma})$; applying the diagonal-production corollary to checked derivability therefore requires a separate representability construction.

3.2 DIAGONAL PRODUCTION

Definition 3.2.1 (Goal-restricted anti-diagonal production). Fix an arbitrary type Code and a supplied operation

$$\text{eval} : \text{Code} \rightarrow \text{Code} \rightarrow \text{Form}. \quad (32)$$

For a fixed goal formula G , the goal-restricted principle represents only the behaviour

$$x \mapsto \text{eval}(x, x) \rightarrow G. \quad (33)$$

We write this principle as

$$\text{Eval}_G(C) \triangleq \exists c \forall x \text{eval}(c, x) \simeq_C \text{eval}(x, x) \rightarrow G. \quad (34)$$

The case $G = \perp$ is the only instance used below. This is a supplied representability premise, not an evaluator constructed from the formula grammar. No quotation operation, coding of formulas, or general completeness condition is assumed.

Remark 3.2.2. Closure equivalence is essential. Replacing $\simeq_C$ by formula equality would produce $B = \neg B$, contradicting Lemma 2.1.2.

Theorem 3.2.3 (Goal-restricted evaluation produces a negation fixed point). Under $\text{Eval}_\perp(C)$,

$$\exists B : \text{Form}, \text{NegFP}_C(B). \quad (35)$$

Proof. Use $\text{Eval}_\perp(C)$ to obtain a code c such that

$$\forall x, \text{eval}(c, x) \simeq_C \text{eval}(x, x) \rightarrow \perp. \quad (36)$$

Set $B \triangleq \text{eval}(c, c)$. Specializing the representing property at $x := c$ gives

$$B \simeq_C B \rightarrow \perp, \quad (37)$$

which is exactly $\text{NegFP}_C(B)$ because $\neg B \triangleq B \rightarrow \perp$. ■

Corollary 3.2.4 (Goal-restricted obstruction to closure-level EXCLUDED MIDDLE). Assume $\text{Eval}_\perp(C)$, $\text{MP}(C)$, and $\text{Cons}(C)$. Then

$$\neg \text{LEM}(C). \quad (38)$$

Proof. Theorem 3.2.3 supplies a formula B with $\text{NegFP}_C(B)$. Corollary 3.1.7 applies at that formula. ■

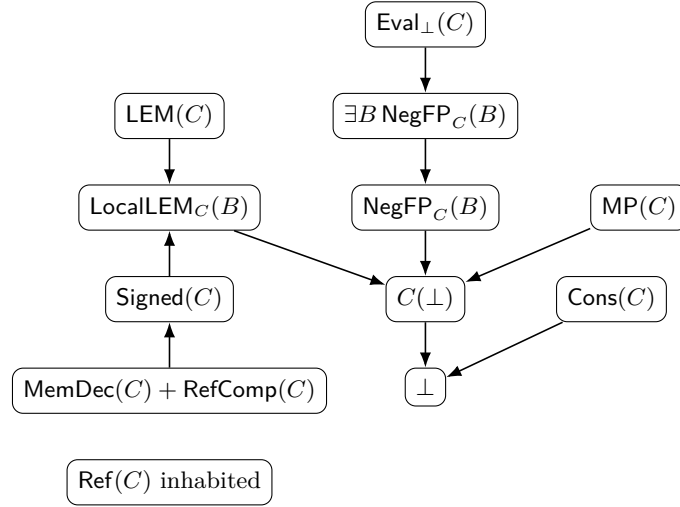


Figure 1. Dependency structure of the obstruction. Goal-restricted production supplies the fixed point; closure-level EXCLUDED MIDDLE and the classifiers supply the local branch; negative-only refutation remains separate.

3.3 FURTHER CLASSIFICATION CONSEQUENCES

Theorem 3.3.1 (Local signed-classification obstruction). For any formula B , the following hypotheses are incompatible:

$$\text{NegFP}_C(B) \wedge \text{MP}(C) \wedge \text{Cons}(C) \wedge \text{Signed}(C) \implies \perp. \quad (39)$$

Proof. Let s be the signed classifier and inspect $s(B)$. A true verdict supplies $C(B)$; a false verdict supplies $C(\neg B)$. Thus $\text{LocalLEM}_C(B)$ holds, and Theorem 3.1.5 applies. ■

Remark 3.3.2. The classifier is global as a function, but the proof reads it only once, at the supplied fixed point B . No uniform bivalence schema is required.

Lemma 3.3.3 (Membership decision plus refutation completeness gives signed classification). If $\text{MemDec}(C)$ and $\text{RefComp}(C)$ hold, then $\text{Signed}(C)$ holds.

Proof. Let m decide membership. Use the same Boolean function as the signed classifier. A true verdict yields $C(A)$ by the membership specification. If $m(A) = \text{ff}$, then $C(A)$ would force $m(A) = \text{tt}$, so $\neg C(A)$; applying $\text{RefComp}(C)$ yields $C(\neg A)$. ■

Theorem 3.3.4 (Local membership-decision obstruction). For any formula B , the following hypotheses are incompatible:

$$\begin{aligned} \text{NegFP}_C(B) \wedge \text{MP}(C) \wedge \text{Cons}(C) \wedge \text{RefComp}(C) \wedge \text{MemDec}(C) \\ \Downarrow \\ \perp. \end{aligned} \tag{40}$$

Proof. Lemma 3.3.3 supplies $\text{Signed}(C)$. Theorem 3.3.1 then applies at B . ■

Theorem 3.3.5 (Refutations are inhabited). For every $C : \text{Form} \rightarrow \text{Prop}$, $\text{Ref}(C)$ is inhabited. Namely, there exists

$$r : \text{Form} \rightarrow \{\text{tt}, \text{ff}\} \quad \text{such that} \quad \forall A : \text{Form}. r(A) = \text{ff}, \tag{41}$$

and the branch soundness condition $r(A) = \text{tt} \rightarrow C(\neg A)$ holds vacuously.

Proof. Set $r(A) := \text{ff}$ for every A . The soundness condition has an empty antecedent, so it is satisfied without invoking either consistency or the diagonal formula B . ■

Remark 3.3.6. All three classification results are separated by what they provide at the fixed point. Signed classification supplies the local branch directly; membership decision supplies it only through $\text{RefComp}(C)$; negative-only refutation supplies no branch and remains inhabited unconditionally. In particular, the theorem does not exclude a membership decider for an arbitrary closure predicate.

3.4 SHARPNESS OF THE LOCAL BOUNDARY

The four hypotheses of the core obstruction are individually necessary at the level of arbitrary closure predicates. This can be seen inside the same closed formula language, without adding atoms or semantics.

Proposition 3.4.1 (Sharpness). For each of $\text{MP}(C)$, $\text{Cons}(C)$, $\text{NegFP}_C(B)$, and $\text{LocalLEM}_C(B)$, there is a closure predicate for which the other three conditions hold while the selected condition fails.

Proof. Use $B := \perp$ and abbreviate

$$T \triangleq \perp \rightarrow \perp, \quad U \triangleq \perp \rightarrow T, \quad V \triangleq T \rightarrow \perp. \tag{42}$$

Thus $T = \neg B$, while $U = B \rightarrow \neg B$ and $V = \neg B \rightarrow B$. The four formulas $\perp, T, U, V$ are syntactically distinct. (i) If consistency is omitted, let C accept every formula. Then $\text{MP}(C)$, $\text{NegFP}_C(B)$, and $\text{LocalLEM}_C(B)$ hold, while

$C(\perp)$ makes $\text{Cons}(C)$ fail. (ii) If local EXCLUDED MIDDLE is omitted, let C accept exactly U and V . This supplies $\text{NegFP}_C(B)$ and remains consistent. MODUS PONENS holds vacuously because the antecedents of the two accepted implications are respectively $\perp$ and T , neither of which is accepted. Hence neither side of $\text{LocalLEM}_C(B)$ holds. (iii) If MODUS PONENS is omitted, let C accept exactly T, U, V . Then $\text{Cons}(C)$, $\text{NegFP}_C(B)$, and $\text{LocalLEM}_C(B)$ hold. But $C(V)$ and $C(T)$ hold while $C(\perp)$ does not, so detachment along $V = T \rightarrow \perp$ fails. (iv) If the negation fixed point is omitted, let C accept exactly T . Then $\text{Cons}(C)$ and $\text{LocalLEM}_C(B)$ hold, and MODUS PONENS is again vacuous because $T = \perp \rightarrow \perp$ is accepted while its antecedent $\perp$ is not. Neither U nor V is accepted, so $\text{NegFP}_C(B)$ fails. ■

4 MECHANIZATION

Mechanization 4.0.1. The mechanization separates the executable syntactic substrate from the abstract closure obstruction.



Figure 2. The M001 substrate and the six L001 contracts underlying the obstruction are formalized in GALLINA, the specification language of the ROCQ proof environment.

M001 provides the closed $\perp/\rightarrow$ syntax, the theorem that no formula equals its own negation, a total Boolean proof checker, executable MP and deduction transformations, an executable refutation-to-negation transformation named `reductio` in the artifact (constructive negation introduction, not a classical principle), syntactic adequacy of checked derivability, and context/theory monotonicity. Its five-contract aggregate `M001_CONTRACT` comprises `SYNTAX`, `ADEQUACY`, `DEDUCTION`, `REDUCTIO`, and `STRUCTURE`, certified by `m001_contract_qed`. Its `AxiomSet` is a caller-supplied Boolean membership function; `FiniteAxiomSet` is the separate finite-data interface. The five executable roots are extracted as `M001_checker.ml/.mli`:

GitHub → Milan-Rosko/Proofcase → M001

At the derivability level, `regulator_theory_checked_derivable_mp_lemma` derives Proposition 2.2.13 from checked proof-script composition. Corollary 3.1.8 therefore applies the abstract obstruction to every induced predicate $C_{R,\Gamma}$ without adding MODUS PONENS as a separate regulator-specific premise. L001 uses only M001’s primitive formula syntax. It defines the closure vocabulary and goal-restricted representability premise abstractly over $C : \text{Form} \rightarrow \text{Prop}$, then certifies the six contracts listed in Table 1. The derived endpoints `branchwise_goal_relative_collapse`, `goal_relative_branch_collapse`, and `fixedpoint_gap`, together with the local-detachment bridge, formalize the refinements above without adding clauses to the aggregate:

GitHub → Milan-Rosko/Proofcase → L001

Theorem	Reference	Endpoint
No formula is its own negation	2.1.2	<code>formula_not_self_negation</code>
Local branch collapse	3.1.3	<code>local_branch_collapse</code>
Core diagonal obstruction	3.1.5	<code>core_diagonal_obstruction</code>
Goal-restricted fixed point	3.2.3	<code>eval_bottom_negfixp</code>
Local signed obstruction	3.3.1	<code>local_signed_obstruction</code>
Local membership obstruction	3.3.4	<code>local_membership_obstruction</code>
Refutation inhabited	3.3.5	<code>closure_refutation_inhabited</code>
L001 aggregate contract	—	<code>l001_contract_qed</code>

Table 1. Mapping between the retained contract statements and the reduced ROCQ endpoints. The first row belongs to M001; the six closure results and aggregate belong to L001.

The goal-relative, branchwise, local-detachment, and fixed-point-gap results are derived L001 theorems rather than additional contract clauses. The paper-level LEM corollaries compose the certified endpoints, and the checked-regulator instance additionally uses the M001 derivability bridge; none enlarges the six-contract L001 boundary. Proposition [3.4.1](#) is likewise a paper-level proof rather than an L001 contract. The M001 and L001 aggregate theorems are `m001_contract_qed` and `l001_contract_qed`. The artifact build is configured to generate assumption reports for both aggregates, and the source contains neither user axioms nor admitted proofs. L001 performs no OCAML extraction because its obstruction contracts live in `Prop`. Its relevant artifact is therefore the assumption report, while M001 separately supplies the executable checker artifact.

5 DISCUSSION

5.1 THE LOCALITY OF THE OBSTRUCTION

The global conclusion is $\neg \text{LEM}(C)$, but the theorem needs only the single branch instance

$$C(B) \vee C(\neg B). \quad (43)$$

The goal-restricted premise represents the one behaviour

$$x \mapsto \text{eval}(x, x) \rightarrow \perp. \quad (44)$$

At its representing code c , self-application gives $B \triangleq \text{eval}(c, c)$ with $B \simeq_C \neg B$. The evaluation operation then has no further role: in either local branch, the collapse uses only the fixed-point implication oriented from that branch and two of the three detachment instances isolated in Theorem [3.1.1](#). Under consistency, Theorem [3.1.4](#) rejects both branches at this particular B ; that external diagnosis is not itself an accepted formula of C .

5.2 WHERE A CLOSURE PREDICATE CAN ESCAPE

Proposition [3.4.1](#) establishes four independent escape routes. A closure predicate may fail to admit a formula B with $B \simeq_C \neg B$; it may admit such

a point but reject the local branch $C(B) \vee C(\neg B)$; it may fail to support MODUS PONENS for the relevant accepted implications; or it may accept $\perp$ and therefore fail the stated consistency condition.

The result asserts only that these four ingredients cannot be combined. It does not privilege one escape route. Closure-level EXCLUDED MIDDLE is ruled out because it supplies the obstructed branch at every formula and therefore at B ; no totality requirement is used anywhere else in the proof.

5.3 COMPARISONS AND OUTLOOK

The goal-restricted representability premise is deliberately narrower than a general theory of evaluation. It names only the anti-diagonal behaviour needed at $\perp$. The fixed point it supplies is not an equality:

$$B \simeq_C \neg B \quad \text{meaning} \quad C(B \rightarrow \neg B) \wedge C(\neg B \rightarrow B). \quad (45)$$

Substructural treatments sharpen the comparison. Roberts' refinement of LAWVERE'S FIXED-POINT THEOREM [Roberts, 2023] shows that diagonal arguments can survive substantial weakening of surrounding structure. Here the branchwise collapse names only three local detachment instances; global modus-ponens closure is a convenient sufficient condition for them. No additional structural rules are posited inside the object language.

The result should also be contrasted with universal-schema presentations of DIAGONALIZATION, such as Yanofsky [2003]. Such presentations treat many paradoxes and limitative theorems as instances of a common diagonal-point pattern. The present theorem belongs to this tradition, but the fixed point is formed at the level of $\simeq_C$. Unlike those presentations, the relevant point here is not syntactic, semantic, or provable equality, but closure equivalence alone:

$$B \simeq_C \neg B. \quad (46)$$

Comparison with the SECOND INCOMPLETENESS THEOREM [Gödel, 1931] and REFLECTION PRINCIPLES [Feferman, 1962] is indirect. Those results concern systems that internalize derivability; the present theorem assumes only a formula-indexed closure predicate and a separately supplied closure-equivalence fixed point.

The same caution applies to computability language. The signed-classification theorem excludes a Boolean function whose negative verdict is already certified by $C(\neg A)$. It is not, merely by renaming, an undecidability theorem for membership in C . The latter conclusion is available here only under the explicit refutation-completeness bridge, and recursion-theoretic readings require a separately fixed effective coding and computation model.

The branchwise collapse already answers part of the one-way question. Under consistency and $\text{MP}(C)$, the mirror $C(\neg\chi \rightarrow \chi)$ excludes $C(\neg\chi)$: that accepted branch would recover $C(\chi)$ and hence $C(\perp)$. What remains open is whether this non-refutability can block coded recovery of the missing direction $\chi \rightarrow \neg\chi$, whose acceptance would complete the fixed point, and whether such opacity can persist through finite mirror iterations.

5.4 SUMMARY

No consistent primitive regulator can combine MODUS PONENS, a closure-equivalence negation fixed point, and closure-level EXCLUDED MIDDLE. At the fixed point, consistency excludes both $C(B)$ and $C(\neg B)$; each attempted branch uses only its corresponding fixed-point direction and two local detachment instances. The supplied goal-restricted evaluation premise produces such a point without a general evaluation structure. Signed classification supplies the same obstructed branch directly; membership decision supplies it only with refutation completeness; negative-only refutation remains inhabited.

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NOTE ON THE THIRD VERSION

This third version brings the exposition into line with the streamlined M001 and L001 developments. It also makes the main argument more transparent: the collapse works for any target formula, each case uses only the implication and detachment steps it needs, and consistency rules out both sides at the fixed point. These refinements are checked by derived L001 theorems; the six certified contracts are unchanged.

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Milan Rosko

ORCID: [0009-0003-1363-7158](https://orcid.org/0009-0003-1363-7158)

Email: hi*atsymbol*milanrosko.com

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